

## GORNA ORYAHOVITSA, Bulgaria

WMO: 155253

Lat: 43.151N

Lon: 25.713E

Elev: 87

StdP: 100.29

Time Zone: 2.00 (EUE)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.8      | -11.1 | -16.1                         | 0.9 | -14.1 | -12.9 | 1.2 | -10.7 | 11.9                  | 4.4  | 10.5 | 4.6  | 0.7                   | 90   | 0.483 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.3                   | 35.1            | 21.4 | 33.2 | 21.3 | 31.8 | 20.9 | 23.7                | 30.5 | 22.8 | 29.4 | 22.0 | 28.5 | 3.0                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 21.9                            | 16.8 | 26.7 | 20.9 | 15.8 | 25.5 | 20.0 | 14.9 | 24.5 | 71.4          | 30.6 | 67.9 | 29.6 | 64.9 | 29.0 | 27.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |     |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |  |
|-------------------|-----|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|--|
| 1%                |     | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |  |
| Min               |     | Max  |                            | Min   | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |  |
| (a)               | (b) | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |
| 9.8               | 8.2 | 6.7  | DB                         | -19.4 | 37.6 | 3.9                | 2.3                                                | -22.3     | 39.3 | -24.6      | 40.6 | -26.8      | 41.9 | -29.7      | 43.5 |  |
|                   |     |      | WB                         | -19.4 | 25.1 | 4.0                | 1.1                                                | -22.3     | 25.9 | -24.6      | 26.5 | -26.9      | 27.1 | -29.8      | 27.9 |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.0   | -0.7  | 2.4   | 7.0   | 12.1  | 16.9  | 21.0  | 23.1  | 22.8  | 17.9  | 12.3  | 7.3   | 1.4  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.13   | 5.71  | 5.83  | 4.36  | 3.62  | 2.96  | 2.89  | 2.56  | 2.61  | 3.37  | 4.16  | 4.50  | 5.25 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1083   | 332   | 217   | 113   | 19    | 1     | 0     | 0     | 0     | 25    | 106   | 270   |      | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2746   | 589   | 447   | 351   | 189   | 61    | 9     | 1     | 2     | 48    | 189   | 333   | 526  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1816   | 1     | 3     | 20    | 81    | 216   | 329   | 406   | 398   | 238   | 97    | 24    | 3    | (10) |
| (11) |                                                                                  | CDD18.3     | 437    | 0     | 0     | 0     | 2     | 18    | 88    | 149   | 141   | 36    | 3     | 1     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 5638   | 0     | 0     | 6     | 67    | 358   | 1064  | 1780  | 1742  | 536   | 82    | 4     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 2430   | 0     | 0     | 0     | 11    | 81    | 428   | 853   | 851   | 189   | 17    | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.4    | 2.2   | 2.5   | 3.0   | 2.8   | 2.5   | 2.3   | 2.5   | 2.1   | 2.2   | 2.0   | 2.2   | 2.1  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 650    | 43    | 42    | 49    | 54    | 78    | 68    | 67    | 47    | 63    | 46    | 41    | 56   | (15) |
| (16) |                                                                                  | PrecMax     | 1125   | 102   | 89    | 152   | 111   | 167   | 138   | 202   | 140   | 221   | 114   | 96    | 97   | (16) |
| (17) |                                                                                  | PrecMin     | 423    | 10    | 6     | 3     | 3     | 23    | 9     | 4     | 4     | 3     | 1     | 2     | 8    | (17) |
| (18) |                                                                                  | PrecStd     | 163    | 21    | 24    | 33    | 28    | 43    | 40    | 53    | 35    | 52    | 29    | 25    | 24   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 16.8  | 20.1  | 24.0  | 28.1  | 30.9  | 35.1  | 38.9  | 37.1  | 33.9  | 29.0  | 23.8  | 18.1 | (19) |
| (20) |                                                                                  | 0.4%        | MCWB   | 11.5  | 12.7  | 13.7  | 17.6  | 19.5  | 21.9  | 21.5  | 20.9  | 19.6  | 18.1  | 16.1  | 12.2 | (20) |
| (21) |                                                                                  | 2%          | DB     | 13.2  | 17.0  | 21.1  | 25.2  | 28.9  | 32.9  | 35.2  | 35.0  | 31.0  | 25.8  | 20.9  | 14.9 | (21) |
| (22) |                                                                                  | 2%          | MCWB   | 9.3   | 11.1  | 12.4  | 16.1  | 18.6  | 22.1  | 21.1  | 21.5  | 18.9  | 16.9  | 15.1  | 10.5 | (22) |
| (23) |                                                                                  | 5%          | DB     | 10.1  | 14.1  | 18.9  | 22.2  | 27.1  | 31.0  | 33.2  | 33.1  | 28.9  | 23.0  | 18.2  | 12.1 | (23) |
| (24) |                                                                                  | 5%          | MCWB   | 7.5   | 9.7   | 11.4  | 14.4  | 18.1  | 21.3  | 21.2  | 21.2  | 18.4  | 16.2  | 13.6  | 8.9  | (24) |
| (25) |                                                                                  | 10%         | DB     | 7.0   | 11.1  | 16.1  | 20.2  | 25.2  | 29.1  | 31.1  | 31.2  | 26.8  | 20.2  | 15.2  | 8.9  | (25) |
| (26) |                                                                                  | 10%         | MCWB   | 5.4   | 7.7   | 10.1  | 13.3  | 17.6  | 20.6  | 20.9  | 20.4  | 18.0  | 15.0  | 11.9  | 7.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.8  | 13.2  | 14.6  | 18.7  | 21.5  | 24.6  | 24.6  | 25.5  | 21.7  | 19.7  | 16.5  | 12.9 | (27) |
| (28) |                                                                                  | 0.4%        | MCDWB  | 16.2  | 18.5  | 20.9  | 26.3  | 27.7  | 31.5  | 31.7  | 32.0  | 28.3  | 26.4  | 22.1  | 17.5 | (28) |
| (29) |                                                                                  | 2%          | WB     | 9.8   | 11.3  | 13.0  | 16.9  | 20.2  | 23.3  | 23.6  | 23.7  | 20.4  | 18.0  | 15.2  | 10.7 | (29) |
| (30) |                                                                                  | 2%          | MCDWB  | 13.3  | 16.2  | 19.7  | 23.9  | 25.9  | 29.9  | 30.6  | 30.1  | 27.2  | 23.3  | 20.2  | 14.1 | (30) |
| (31) |                                                                                  | 5%          | WB     | 7.6   | 9.9   | 11.8  | 15.4  | 19.2  | 22.4  | 22.7  | 22.6  | 19.5  | 16.6  | 13.9  | 9.2  | (31) |
| (32) |                                                                                  | 5%          | MCDWB  | 9.7   | 14.3  | 17.9  | 20.8  | 25.3  | 28.7  | 29.4  | 29.1  | 26.3  | 21.8  | 17.7  | 11.9 | (32) |
| (33) |                                                                                  | 10%         | WB     | 5.3   | 8.0   | 10.5  | 14.1  | 18.2  | 21.4  | 21.9  | 21.6  | 18.6  | 15.4  | 12.4  | 7.2  | (33) |
| (34) |                                                                                  | 10%         | MCDWB  | 6.7   | 10.7  | 15.6  | 19.3  | 23.8  | 27.4  | 28.4  | 28.4  | 24.9  | 19.6  | 15.3  | 8.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 9.0   | 10.9  | 12.3  | 13.3  | 13.7  | 13.6  | 14.3  | 15.2  | 14.0  | 12.9  | 10.1  | 9.1  | (35) |
| (36) |                                                                                  | 5% DB       | MCDDBR | 14.4  | 16.7  | 18.5  | 17.5  | 17.6  | 16.8  | 18.4  | 18.5  | 18.8  | 18.2  | 15.6  | 14.9 | (36) |
| (37) |                                                                                  |             | MCWBR  | 10.4  | 10.8  | 10.2  | 9.5   | 8.7   | 7.1   | 6.7   | 6.7   | 7.8   | 10.1  | 9.9   | 10.6 | (37) |
| (38) |                                                                                  |             | MCDDBR | 13.6  | 16.0  | 16.4  | 15.4  | 15.1  | 15.1  | 14.7  | 15.2  | 15.5  | 16.0  | 14.1  | 13.5 | (38) |
| (39) |                                                                                  | 5% WB       | MCWBR  | 10.0  | 10.6  | 9.4   | 8.8   | 8.3   | 7.3   | 6.6   | 6.7   | 7.4   | 9.3   | 9.2   | 9.8  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.350  | 0.384 | 0.429 | 0.487 | 0.461 | 0.467 | 0.474 | 0.487 | 0.453 | 0.410 | 0.376 | 0.352 | (40) |      |
| (41) |                                                                                  | taud        | 2.273  | 2.192 | 2.114 | 1.967 | 2.114 | 2.141 | 2.131 | 2.088 | 2.147 | 2.246 | 2.293 | 2.279 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 758    | 791   | 801   | 782   | 818   | 813   | 801   | 774   | 769   | 753   | 721   | 714   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 89     | 114   | 139   | 174   | 154   | 151   | 151   | 152   | 133   | 107   | 87    | 81    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.52   | 2.31  | 3.40  | 4.68  | 5.80  | 6.35  | 6.49  | 5.84  | 4.27  | 2.78  | 1.66  | 1.26  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.22   | 0.29  | 0.31  | 0.50  | 0.46  | 0.46  | 0.46  | 0.34  | 0.33  | 0.30  | 0.21  | 0.17  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (41 neighbors) | +0.44 | N/A     | N/A    | N/A     | +0.43 | N/A   | N/A         | -128    | +84     | N/A     |

Nomenclature: See separate page